THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 1

Chapter 12: **How Nature Works in Harmony** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. An ecosystem consists of: 1

(a) living and non-living components interacting with each other
(b) only humans
(c) only soil and water
(d) only plants and animals

2. A series of organisms in which each one is eaten by the next is a: 1

(a) food web **(b)** ecosystem **(c)** habitat **(d)** food chain

3. The process of conversion of dead matter into simple substances by decomposers is: 1

(a) decomposition **(b)** photosynthesis **(c)** respiration **(d)** pollination

4. Which of the following is an abiotic component? 1

(a) grass **(b)** sunlight **(c)** earthworm **(d)** fungi

5. Which of the following is a correct food chain? 1

(a) snake, frog, grass **(b)** grass, grasshopper, frog, snake
(c) frog, grass, snake **(d)** grasshopper, snake, grass

6. Which gas do green plants take in during photosynthesis? 1

(a) nitrogen **(b)** carbon dioxide **(c)** hydrogen **(d)** oxygen

7. Which of the following is a biotic component? 1

(a) air **(b)** water **(c)** bacteria **(d)** soil particles

8. In a food chain, the first trophic level is occupied by: 1

(a) decomposers **(b)** carnivores **(c)** producers **(d)** herbivores

9. Which gas do plants release during photosynthesis? 1

(a) carbon dioxide **(b)** methane **(c)** oxygen **(d)** nitrogen

10. Organisms that make their own food are called: 1

(a) consumers **(b)** parasites **(c)** producers **(d)** decomposers

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. What is an ecosystem? Give two examples. 2

12. What do decomposers do? Name two. 2

13. What is biodiversity? Why is it important? 2

14. Differentiate between biotic and abiotic components with two examples each. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Draw in words a pond ecosystem and name its biotic and abiotic components. 3

16. Explain nitrogen fixation and its importance. 3

17. Explain the importance of decomposers in nature. 3

OR

Explain the role of producers, consumers and decomposers in an ecosystem. 3

18. State three effects of deforestation. 3

OR

State three characteristics of a forest ecosystem. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe the components of an ecosystem with examples and show how they are interdependent. 5

OR

Explain how the removal of one species affects an ecosystem. Give an example of vultures or tigers. 5

20. Describe food chains and food webs with two examples and explain why a food web is more stable than a food chain. 5

OR

Describe how plants and animals help each other: pollination, seed dispersal, symbiosis with two examples. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND